



MATHEMATICS

THE KENYA NATIONAL EXAMINATION COUNCIL KCPE 2015

Time: 2 hours

1. What is 6090428 written in words?
 - A. Sixty million ninety thousand four hundred and twenty eight.
 - B. Six million ninety thousand four hundred and twenty eight.
 - C. Sixty million nine thousand four hundred and twenty eight.
 - D. Six million nine thousand four hundred and twenty eight.
2. What is the value of $(549 - 243 \div 27) \div 27$
 - A. 540
 - B. $118\frac{2}{3}$
 - c. 102%
 - D. 20
3. What is the value of $\frac{5}{6} + \frac{7}{4} + \frac{7}{12} - \frac{1}{3}$
 - A. $\frac{3}{1/2}$
 - B. $\frac{1}{24/48}$
 - c. $\frac{4}{2/21}$
 - D. $\frac{10}{1/3}$
4. What is the value of $0.204 + (0.02 \times 0.17)$?



A. 0.06

B. 0.6

C. 6

D. 60

5. What is 2806198 rounded off to the nearest tens?

A. 2806190

B. 2806210

C. 2806208

D. 2806200

6. What is the measure of the reflex angle Working Space PQR drawn below? ' A.315

B . 225°

C. 135°

D. 45°

7. What is the difference in the value of the square root of $\frac{2}{14/25}$ and the square of $\frac{2}{5}$?

A $\frac{2}{2/5}$

B. $\frac{1}{19/25}$

C. $\frac{1}{11/25}$

D.. $\frac{11}{5}$

8: In a kiosk the prices of food were shown as follows:

Items 'PRICE of Items in Shillings

Tea20

Andazi10

seefszew80



Fish 75

Ugali25

Chapati20

Githeri.....4

Three boys ordered beef stew, ugali and a cup of tea each while eight girls ordered fish and chapati each. What was their total bill?

A. sh 220

B. sh 1075

C. sh 1135

D. sh 1295

9. What is the simplified form of the expression $3(3x - y + 1) + 2(3y - x + 4)$?

A. $8x + 5y + 5$

B. $11x + 9y + 11$

C. $7x + 3y + 11$

D. $2x + 2y + 10$

10. A strip of metal was painted in three different colours. Two fifths of the strip was painted white, $\frac{1}{4}$ of the strip was painted green and the remaining part was painted black. Which is the correct order of writing the fractions of the strip, painted from the smallest to the largest?

A $\frac{1}{8}, \frac{2}{5}, \frac{19}{40}$

B- $\frac{19}{40}, \frac{2}{5}, \frac{1}{8}$

C- $\frac{19}{40}, \frac{1}{8}, \frac{2}{5}$

D- $\frac{2}{5}, \frac{1}{8}, \frac{19}{40}$

11. Mwakio fenced a rectangular piece of land measuring 15 m and 31 m. He used six strands of wire, leaving a space of 5 m for the gate. What was the total length of the wire used?



A. 552 m

B. 522 m

C. 547 m

D. 87 M

12. The price of an item was reduced by sh 360. This represented a 20% discount. What was the price of the item after the discount?

A. sh 450

B. sh 1440

C. sh 1800

D. sh 2 160

13 What is the next number in the pattern 5, 13, 25, 41, 61,?

A. 102

B. 85

C. 81

D. 65

14. What is the value of $U S(T-R)^2/(T+R)^2$ if $S=3, R=4, T=9$?

A. $1 \frac{2}{13}$;

B. $1/2/17$

C. 3

D. 9

15. The figure below is an incomplete parallelogram KLMN. Complete the parallelogram.

16 What is the length of the diagonal KM in centimetres'?

A. 2.5



B. 4.0

C. 5.7

D. 3.5

17 The marked price of a T.V set is sh 20000. A 20% discount is allowed when buying the T.V. on cash. On hire purchase terms, only 12 monthly installments of sh 2000 each is paid. How much more does it cost to buy the T.V. set on hire purchase terms than cash?

A. sh 24 000

B. sh 16 000

C. sh 8 000

D. sh 4 000

18. Ngenya refuels his car after every two days, Kerubo refuels hers after every four days while Masai refuels his after every eight days. If they all refuelled their cars on 3rd March 2013, on which date did they refuel their cars together again?

A. 17th March

B. 11th March

C. 7th March

D. 5th March

19 Pupils contributed some money to help needy people. They bought twenty four 2-kg packets of flour, thirty six 1-kg packets of flour and a fifty kilogram bag of sugar. The flour was packed in 500g packets and the sugar in 250 g packets. How many people got both a packet of flour and a packet of sugar?

A. 368

B. 200

C. 168

D. 120



20 A sales girl earns a salary of sh 3 000 per month plus a commission of 5% on the value of goods she sells above sh 10000. In a certain month, she earned a total of sh 7000. What was the value of goods she sold that month?

- A. sh 70 000
- B. sh 80 000
- C. sh 90 000
- D. sh 140 000

21. The table below shows arrival and departure times for buses from a certain company serving route P to T.

TownArrival Time Departure Time

P 7.30 a.m.

Q..... 8.30 a.m.9.00 a.m.

R..... 11.30 a.m..... 11.45 a.m.

S1.45 p.m.1.55 p.m.

T4.15 p.m.

How long does a bus take to travel from town Q to town S'?

- A. 4 h 45 min
- B. 4 h 55 min
- C. 5 h 15 min
- D 5 h 25 min

21 The volume of Water in a pond is 1084 m³. What is the amount of Water in litres?

- A. 10840000
- B. 1084000
- C. 108400
- D. 10840



22. In the triangle JKL below, construct the Working SPICE perpendicular bisectors of lines JL and KL to meet at point M. Join ML.

WHAT is the length of line ML?

.0575?’

A 1.2 cm

B 2.5 cm

C 8.0 cm

D. 4.8cm

23. The of five numbers is 9. Four of the numbers are 5, 10, 5 and 14. What is the median of the five numbers?

A. 5

B. 9

C. 10

D 11

24. The price of an item increased from sh 400 to sh 500. What was the percentage increase?

A. 125%

B. 80%

C. 25%

D. 20%

25. The pie-chart below shows the angles of Working Space the sectors representing the different types of Fruits sold by Mwanaisha.

A.30

B.70

C.120



D.150

26. The area of the curved surface of a cylinder is 1320 cm^2 . The height of the cylinder is 10 cm. What is the radius of the cylinder in centimetres? (Take $\pi = \frac{22}{7}$)

A 30

B 70

C 120

D 150

27. What is the value of n in the inequality $14n + 9 < 41 - 6n$

A $n < \frac{6}{1/4}$

B $n < 4$

C $n < \frac{2}{1/2}$

D $n < \frac{1}{2/5}$

28. The table below shows the number of patients who attended a clinic on certain days of the week. The number of male patients on Friday and that of children on Wednesday are not recorded.

Days Monday Wednesday Friday

Patients Male 1 74 56

Female 11 0 59

29. A total of 623 patients attended the clinic that week. The number of patients who attended the clinic on Wednesday was 226. How many more children than male patients attended the clinic that week?

A. 19

B. 23

C. 37



D. 56

29. What is the difference in value between the largest and the smallest 6-digit number formed using the digits 3, 0, 4, 5, 2, 9?

A. 954320

B. 203459

C. 930861

D. 750861

30. Akello and Muhu shared some oranges in the ratio 4:5 respectively. Muhu received 3 more oranges than Akello. How many oranges did Akello get?

A. 9

B. 12

C. 15

D. 27

31. A mother shared part of her money to three children Senda, Mila and Kilo. Mila received $\frac{1}{3}$ while Kilo received $\frac{1}{4}$ of the money. Senda received $\frac{1}{6}$ of the remainder. What fraction of the money was shared out?

A. $\frac{3}{4}$

B. $\frac{1}{2}$

C. $\frac{1}{3}$

D. $\frac{1}{4}$

32 Which one of the following sets of measurements will form a right angled triangle when constructed?

A. 3 cm, 5 cm, 6 cm

B. 4 cm, 6 cm, 15 cm

C. 8 cm, 12-cm, 15 cm



D. 6 cm, 8 cm, 10 cm

33 How many blocks are used to make the stack drawn below?

A. 60

B. 55

C. 53

D. 45

34 An aeroplane reached its destination on a Wednesday at 01 15h. The flight had taken 4% hours. On what day and at what time in a.m./p.m. system did the plane start its flight?

A. Tuesday 8.45 a.m.

B. Tuesday 8.45 p.m.

C. Wednesday 8.45 a.m.

D. Wednesday 8.45 p.m.

35 A piece of land in the shape of a trapezium is drawn on a map using the scale 1:500. The parallel sides are 5 cm and 2 cm, while the perpendicular distance between parallel sides is 4 cm. What is the actual area of the plot in ml?

A. 35000

B. 3500

C. 350

D. 35

36. In the figure below PQ is parallel to RS. UT and VW are straight lines. What is the size of the angle marked x° ?

A. 44°

B. 73° ,

C. 107°

D. 117°



37. In a certain country, the population of goats is 51907. The number of female goats is 453 more than that of male goats. The number of male goats is 18549 and there are kids. What is the number of kids?

A. 14356

B. 15262

C. 19002

D. 32905

38. The table below shows the postal charges, in Working Space shillings, on small packets in a certain year.

A fiba from Mombasa sent the following packets:

A 900g packet to Uganda

A 5 kg packet to Ghana

A 251 g packet to China

How much money did he pay altogether?

A. sh3091

B. sh2884 C . sh2881

D. sh1831

39. Six men can complete digging a shamba in 8 days. How many days will it take four men, working at the same rate, to complete digging the shamba?

A. 12

B. $5\frac{1}{3}$

C. 4

D. 3

40. The figure below is made up of squares. How many squares are there altogether?

A. 14



B. 13

C.10

D.9

41 The prices of items sold in a shop were as follows:

1kg sugar @ sh 120 _

1 kg rice @ sh 160

1 kg cookingfat @ sh 144

1 loaf of bread @ sh 45

42 Subira bought 2 kg of sugar, 1 kg of rice, %kg of cooking fat and 3 loaves of bread. She gave the shopkeeper a sh 1 000 note. What balance did she get?

A. sh 321

B. sh 393

. C. sh 531

D. sh 607

42 The diagram below represents a solid cube of side 6 cm from which a cylinder of diameter 2.8 cm has been removed.

What is the volumezgf the remaining solid in cm'? (Take 1: = —)

A. 36.96 7

B. 68.16

C. 179.04

D. 2 1 6.0

43. A plot of land is in the shape of a rhombus of side 200 m. The length of one of the diagonals is 240 m. What is the area of the plot in hectares?

A. 3.84

B. 4



C. 4.8

D. 7.68

44. Ndemo bought $2r$ oranges while Maua Working Space bought $4s$ oranges. Haifa bought 2“ oranges more than a half the total number ‘ of oranges bought by both Nderno and Maua. How many oranges did they buy altogether?

A. $3r + 6s + 2$,

B. $4r + 8s + 2$,

C. $6r + 12s + 2$

D. $3r + 6s — 2$

45. The graph below shows joumeys of Musa and Serah. Musa travelled from P to Q while Serah travelled from Q to P.

What was the difference intheir average speeds?

A. 0

B. 10

C. 70

D. 80

46 On a certain day a miller prepared 63 kilograms of uji flour and 286 kilograms of ugali flour. The miller charged sh 7 per kilogram of uji flour and sh 4 per kilogram of ugali flour. The millconsumed 7 litres of fuel costing sh 108 per litre. The miller also paid sh 250 as wages and sh 300 as loan repayment.

What was the miller’s profit that day?

A. i sh 279

B. sh 579

C. sh 927

D. sh 948 .

47 Mundia bought 90 bananas for sh 450. After selling all the bananas he made a profit of 60%. What was the selling price per banana?



- A. sh 5
- B. sh 8
- C. sh 270
- D. sh 720

48 Ngoko deposited sh 120000 in a bank that gave compound interest at the rate of 6% per annum. What was the amount in the bank after 2 years?

- A. sh 134 832
- B. sh 134400
- C. sh 127 200
- D. sh 14832 5

49 Senteu left home at 6.00 a.m. for school which is 8 km away. He cycled at a speed of 6km/h. After cycling for 4km the bicycle had a puncture which took him 15 minutes to repair. He continued cycling at the same speed. At what time did he arrive at school?

- A. 6.40 a.m.
- B. 6.55 a.m.
- C. 7.20 a.m.
- D. 7.35 am.

50. In the triangle EFG shown below, 'Working Space construct a circle that touches the sides of $\triangle$ the triangle.

What is the radius of the circle, in \ centimetres?

- A. 2.5
- B. 5.0
- c. 5.6
- D. 11.2